Table 1: Absolute trajectory error (ATE, cm) on RPG for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value, so each column is a *composite upper bound* over all published results rather than any single reported configuration: no baseline is understated, and no one configuration ever achieved these numbers simultaneously. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code. FAIL marks a cell a source explicitly reports as a failure, which is a measured outcome rather than missing data. Every ES-PTAM, ESIO, ESVIO value is second-hand: its own paper is not the source of any cell here.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [2]	ESIO [2]	DEIO [4]	ES-PTAM [3]
rpg_box	1.92[a]	4.31[b]	4.41[b]	11.38[b]	N/R	4.06[b]
rpg_monitor	0.91[a]	2.31[b]	3.48[b]	7.87[b]	N/R	2.34[b]
rpg_bin	1.16[a]	2.27[b]	2.28[b]	7.08[b]	N/R	2.57[b]
rpg_desk	1.46[a]	1.57[b]	2.03[b]	3.16[b]	N/R	2.84[b]
rpg_reader	3.78[a]	2.68[b]	N/R	FAIL[b]	N/R	N/R

Sources: [a][6], [b][5].

References

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[4] Weipeng Guan, Fuling Lin, Peiyu Chen, and Peng Lu. DEIO: Deep event inertial odometry. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. arXiv:2411.03928.

[5] Junkai Niu, Sheng Zhong, Xiuyuan Lu, Shaojie Shen, Guillermo Gallego, and Yi Zhou. ESVO2: Direct visual-inertial odometry with stereo event cameras. *IEEE Transactions on Robotics*, 2025. arXiv:2410.09374.

[6] Junkai Zhong, Junkai Niu, and Yi Zhou. Stereo deep event visual odometry. *IEEE Robotics and Automation Letters*, 2025. arXiv:2509.08235.

Table 2: Absolute trajectory error (ATE, cm) on MVSEC for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value, so each column is a *composite upper bound* over all published results rather than any single reported configuration: no baseline is understated, and no one configuration ever achieved these numbers simultaneously. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code. FAIL marks a cell a source explicitly reports as a failure, which is a measured outcome rather than missing data. Every ES-PTAM, ESIO, ESVIO value is second-hand: its own paper is not the source of any cell here.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [2]	ESIO [2]	DEIO [4]	ES-PTAM [3]
indoor_flying1	3.76[a]	7.63[b]	9.63[b]	820.4[b]	7.52[a]	15.02[b]
indoor_flying2	8.00[a]	10.05[b]	N/R	417.9[b]	6.96[a]	N/R
indoor_flying3	5.57[a]	7.35[b]	8.06[b]	FAIL[b]	29.75[a]	N/R
indoor_flying4	4.57[a]	5.59[b]	N/R	173.5[b]	12.71[a]	N/R

Sources: [a] [6], [b] [5].

Sequence caveat. Sources evaluate the *edited* MVSEC bags (`indoor_flying*_data_edited`), and further restrict scoring to a sub-window of those: the trajectories released with [5] span 25.8, 24.2, 25.0 and 5.9 s against ground truth of 70.3, 84.9, 94.0 and 19.8 s respectively. Numbers here are therefore ≈26–37% windows, not whole sequences, and are not comparable with a full-sequence run.

Table 3: Absolute trajectory error (ATE, cm) on DSEC for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value, so each column is a *composite upper bound* over all published results rather than any single reported configuration: no baseline is understated, and no one configuration ever achieved these numbers simultaneously. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code. FAIL marks a cell a source explicitly reports as a failure, which is a measured outcome rather than missing data. Every ES-PTAM, ESIO, ESVIO value is second-hand: its own paper is not the source of any cell here.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [2]	ESIO [2]	DEIO [4]	ES-PTAM [3]
city04_a	96.86[a]	56.17[b]	201.5[b]	543.5[c]	80.60[c]	131.6[c]
city04_b	17.65[a]	73.83[b]	48.33[b]	295.1[c]	35.40[c]	29.02[c]
city04_c	481.6[a]	508.7[b]	1400.8[b]	896.2[c]	413.8[c]	1184.4[c]
city04_d	406.2[a]	546.6[b]	921.7[c]	2977.0[c]	207.6[c]	1053.9[c]
city09_a	197.0[a]	1183.0[a]	328.5[a]	FAIL[a]	N/R	FAIL[a]
city09_b	267.8[a]	87.83[b]	3481.5[a]	2887.8[a]	N/R	195.1[b]
city09_c	931.1[a]	1648.3[a]	8043.5[a]	FAIL[a]	N/R	3673.8[a]
city09_d	564.3[a†]	1920.4[a]	1677.1[a]	1766.2[a]	N/R	FAIL[a]
city09_e	246.3[a]	1075.1[a]	438.2[a]	4201.0[a]	N/R	1480.8[a]
city11_a	47.56[a]	48.77[b]	406.1[b]	107.4[b]	N/R	117.9[a]
city11_b	282.1[a]	441.8[b]	N/R	300.1[b]	N/R	FAIL[a]

Sources: [a] [6], [b] [5], [c] [4].

[†] The source reports two different values for this cell: 625.81 cm in its Table II and 564.33 cm in its Table III. We quote the lower under the best-published-value rule.

Table 4: Absolute trajectory error (ATE, cm) on VECtor for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value, so each column is a *composite upper bound* over all published results rather than any single reported configuration: no baseline is understated, and no one configuration ever achieved these numbers simultaneously. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code. FAIL marks a cell a source explicitly reports as a failure, which is a measured outcome rather than missing data. Every ES-PTAM, ESIO, ESVIO value is second-hand: its own paper is not the source of any cell here.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [2]	ESIO [2]	DEIO [4]	ES-PTAM [3]
corner_slow	1.06[a]	2.15[b]	1.43[a]	2.67[b]	10.70[a]	N/R
robot_normal	3.16[a]	4.81[b]	4.95[a]	5.17[b]	N/R	N/R
desk_normal	6.48[a]	16.47[b]	6.00[a]	FAIL[b]	254.8[a]	N/R
sofa_normal	7.18[a]	40.28[b]	5.06[a]	43.94[b]	N/R	N/R
hdr_normal	6.71[a]	13.53[b]	1.96[a]	27.85[b]	N/R	N/R
corridors_dolly	196.5[a]	FAIL[a]	92.04[a]	FAIL[a]	492.6[d]	N/R
units_dolly	444.5[a]	FAIL[a]	872.1[a]	FAIL[a]	826.4[d]	N/R

Sources: [a][6], [b][5], [d][1].

Table 5: Absolute trajectory error (ATE, cm) on TUM-VIE for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value, so each column is a *composite upper bound* over all published results rather than any single reported configuration: no baseline is understated, and no one configuration ever achieved these numbers simultaneously. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code. FAIL marks a cell a source explicitly reports as a failure, which is a measured outcome rather than missing data. Every ES-PTAM, ESIO, ESVIO value is second-hand: its own paper is not the source of any cell here.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [2]	ESIO [2]	DEIO [4]	ES-PTAM [3]
tumvie_1d_trans	1.05[a]	3.33[b]	N/R	FAIL[b]	1.85[a]	1.05[b]
tumvie_3d_trans	1.26[a]	7.26[b]	N/R	FAIL[b]	1.28[a]	8.53[b]
tumvie_6dof	1.69[a]	3.21[b]	N/R	FAIL[b]	1.45[a]	10.25[b]
tumvie_desk	2.44[a]	6.16[b]	N/R	FAIL[b]	1.49[a]	2.50[b]
tumvie_desk2	1.79[a]	4.02[b]	N/R	FAIL[b]	1.45[a]	7.20[b]

Sources: [a][6], [b][5].

Table 6: Absolute trajectory error (ATE, cm) on Other for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value, so each column is a *composite upper bound* over all published results rather than any single reported configuration: no baseline is understated, and no one configuration ever achieved these numbers simultaneously. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code. FAIL marks a cell a source explicitly reports as a failure, which is a measured outcome rather than missing data. Every ES-PTAM, ESIO, ESVIO value is second-hand: its own paper is not the source of any cell here.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [2]	ESIO [2]	DEIO [4]	ES-PTAM [3]
hnu_campus	151.8[a]	181.8[a]	N/R	1420.5[a]	N/R	N/R
drone_fast	13.29[a]	FAIL[a]	N/R	154.0[a]	N/R	N/R

Sources: [a][6].